ЛР 3. Библиотеки программы

Система: macOS, ARM64. Команды выполнены в терминале VS Code.

1. После сборки программа вывела ./libs: sqrt(2) = 1.414214.

2. Команда otool -L libs показала одну зависимость: /usr/lib/libSystem.B.dylib. Математические функции входят в libSystem, поэтому отдельной libm в списке нет.

3. Запускатель Python связан с @executable_path/../Python3. В этой библиотеке находится интерпретатор.

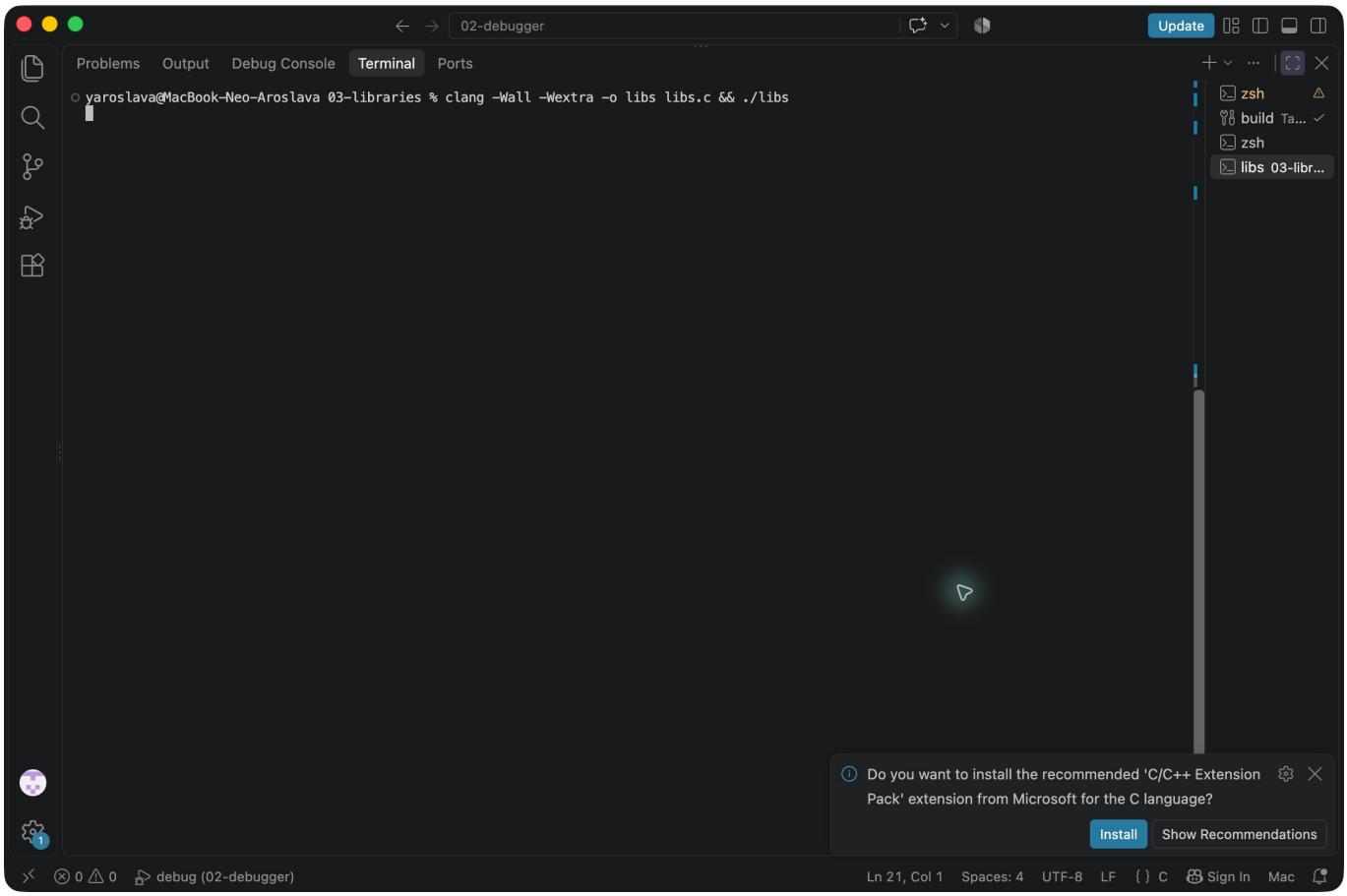
4. PID процесса Python: 29736.

5. До загрузки vmmap показала 172 уникальных пути к файлам .dylib.

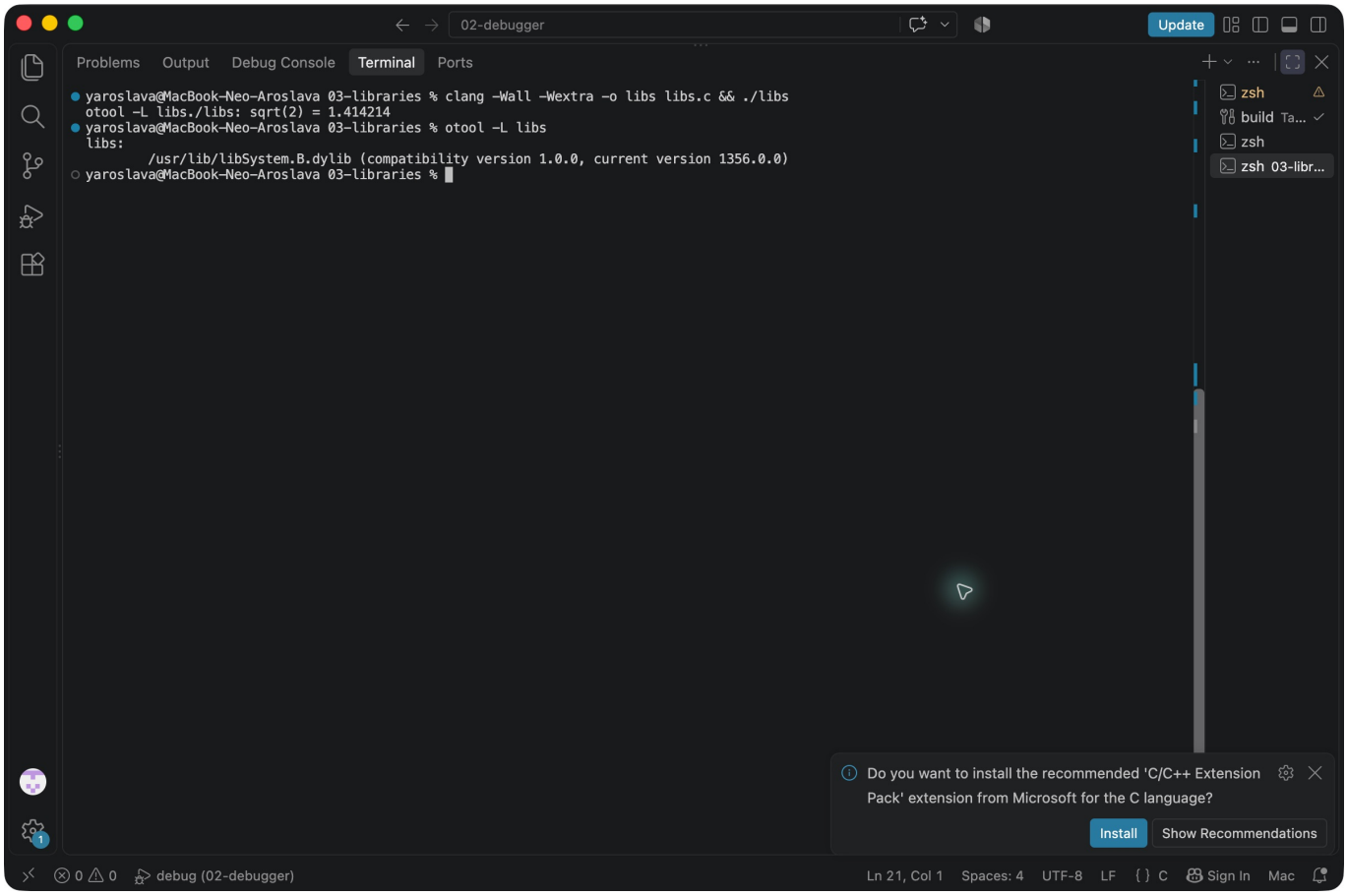
6. ctypes.CDLL загрузила libpcap.dylib. Функция pcap_lib_version вернула libpcap version 1.10.1.

7. После загрузки в том же процессе стало 174 библиотеки. Добавились /usr/lib/libffi.dylib и /usr/lib/libpcap.A.dylib.

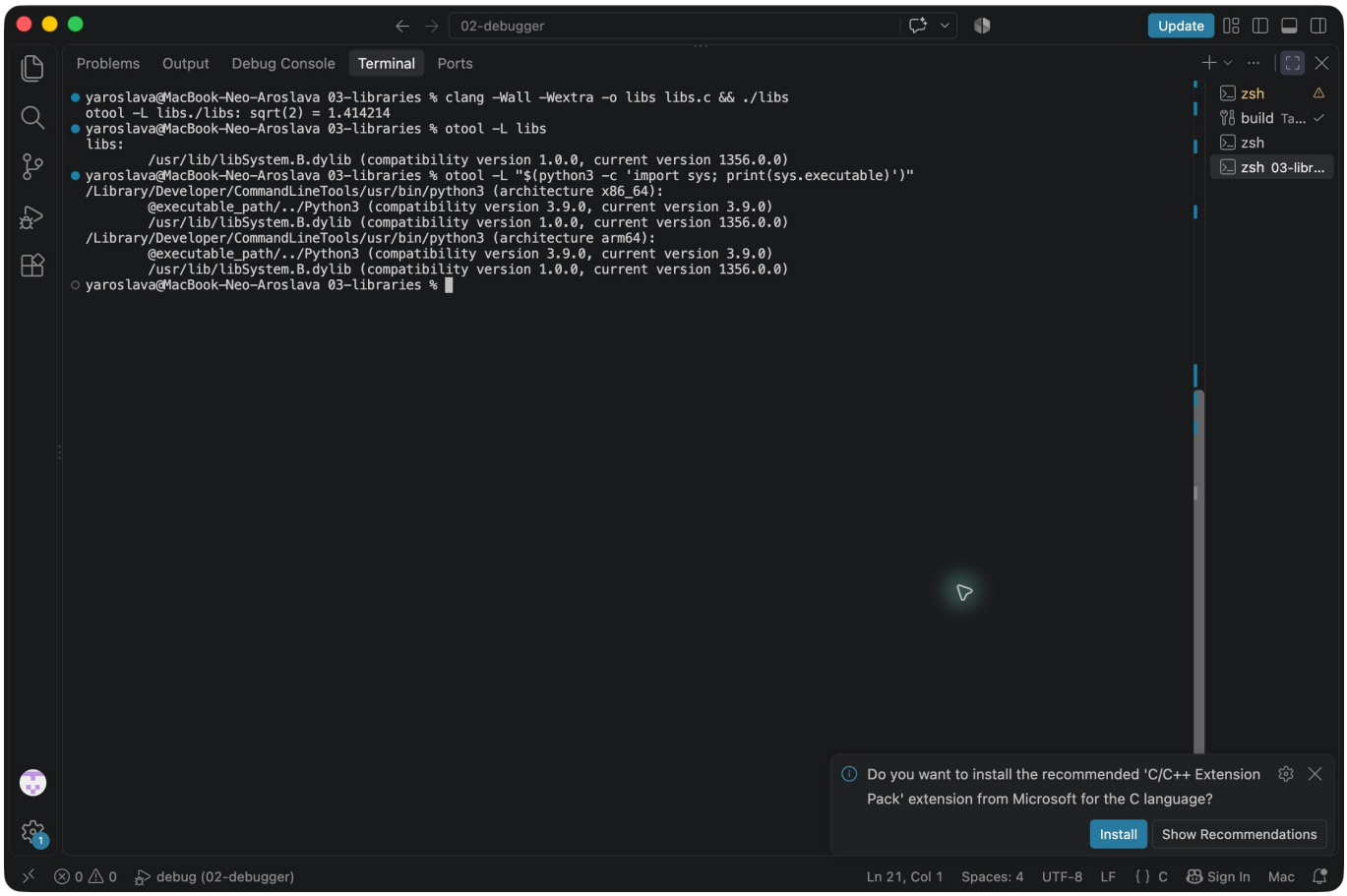
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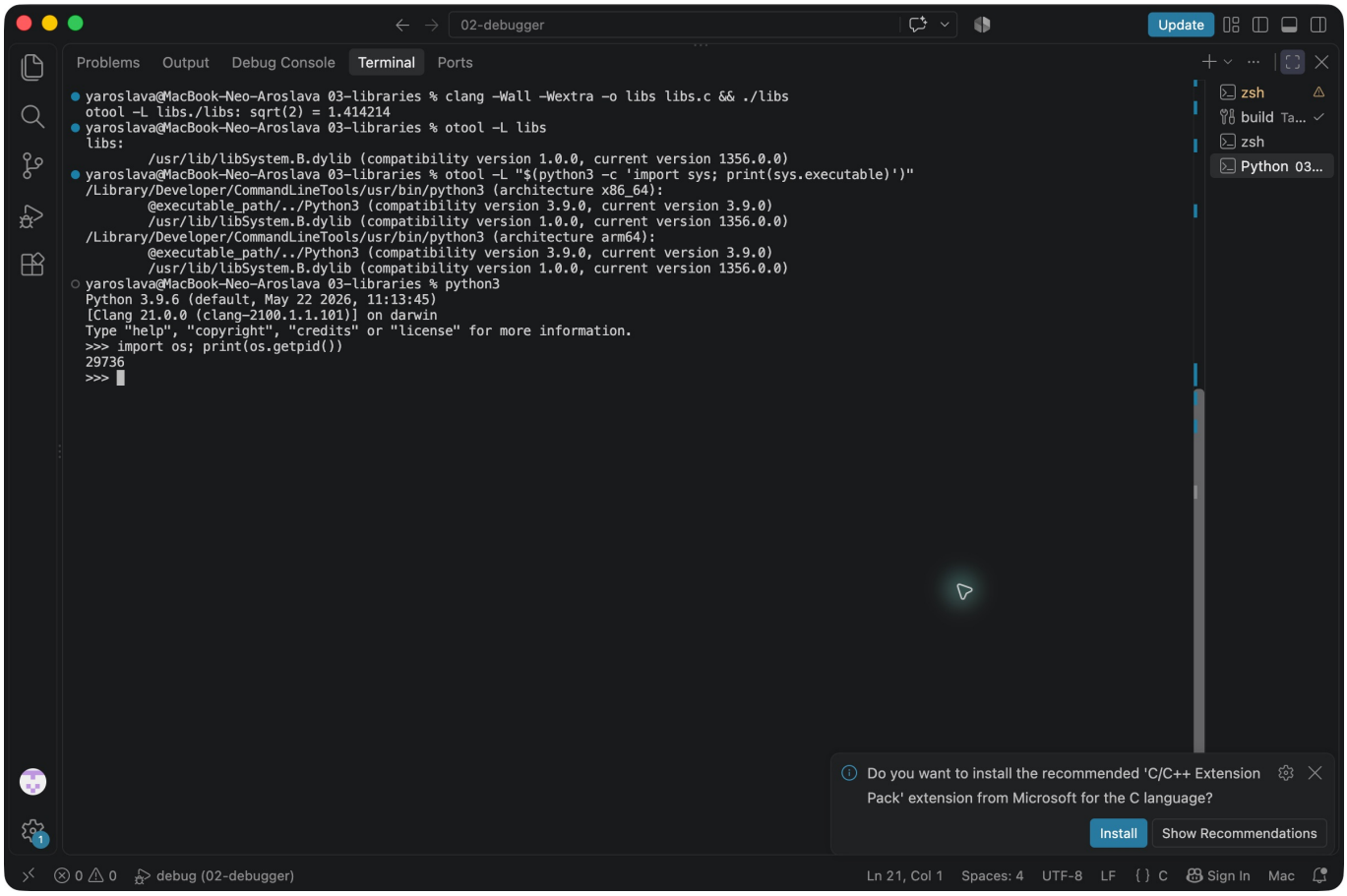
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otool -L libs./libs: sqrt(2) = 1.414214
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libs:
/usr/lib/libSystem.B.dylib (compatibility version 1.0.0, current version 1356.0.0)
yaroslava@MacBook-Neo-Aroslava 03-libraries %
zsh
build Ta...
zsh
zsh 03-libr...
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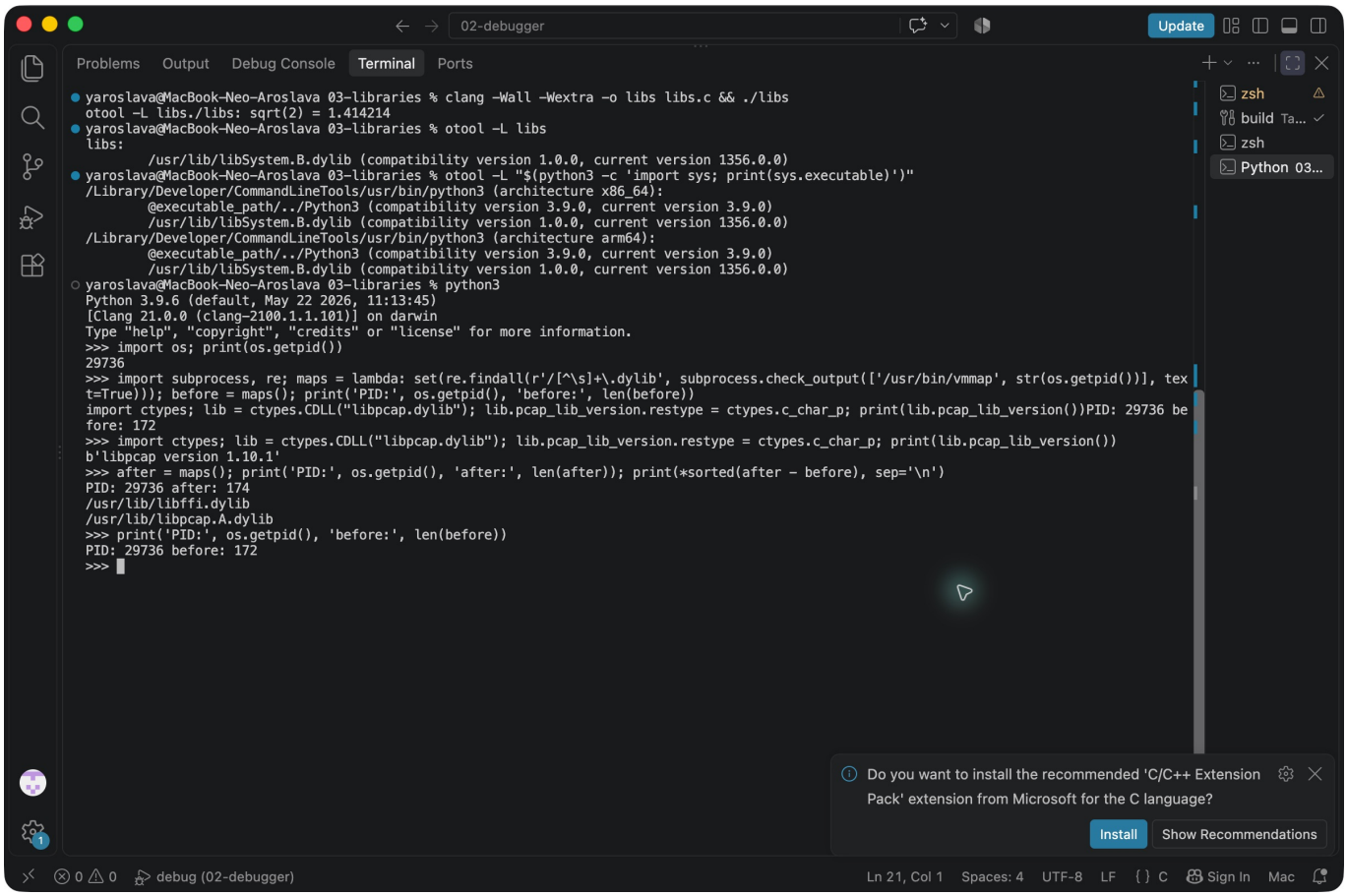
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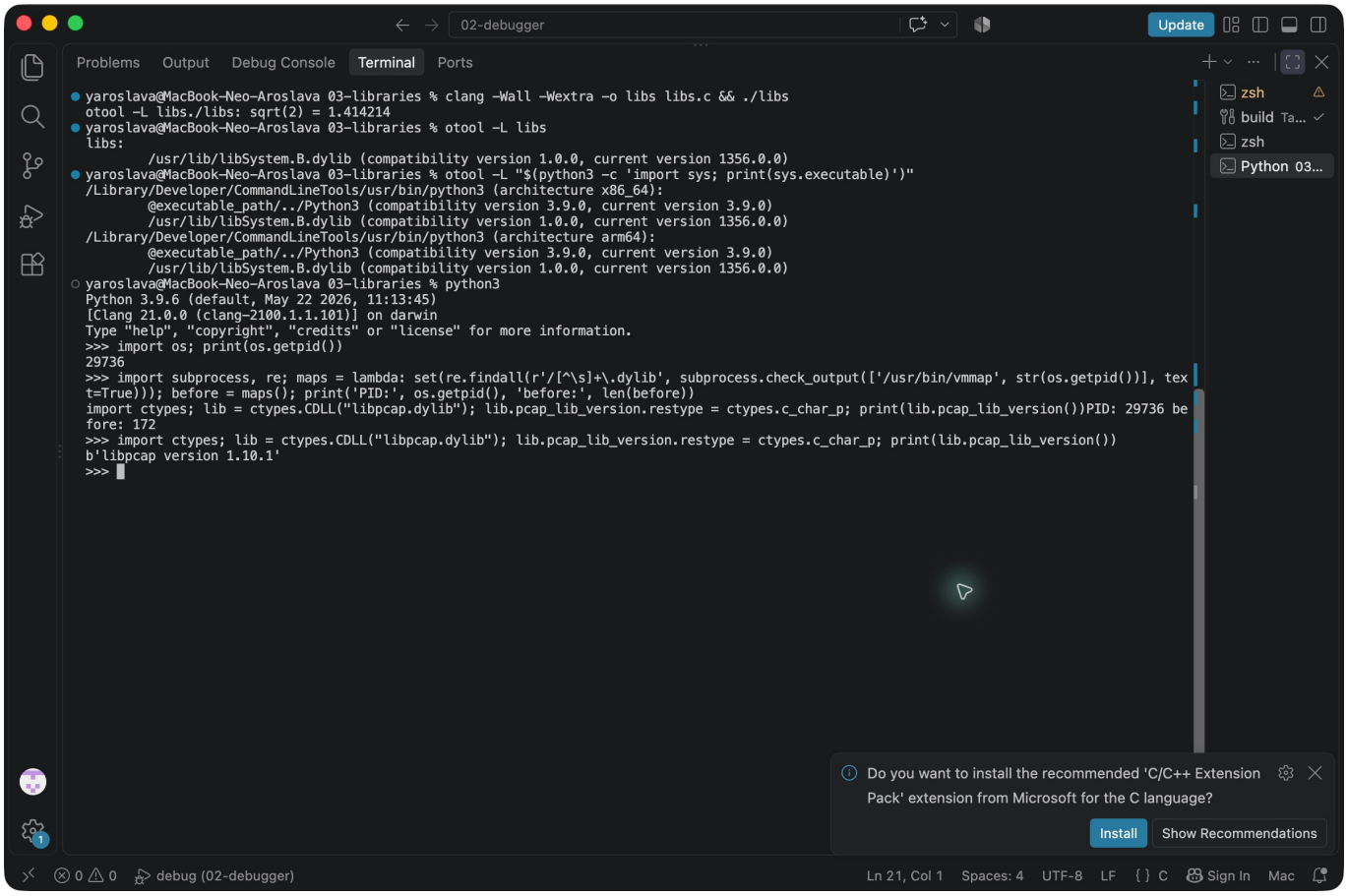
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yaroslava@MacBook-Neo-Aroslava 03-libraries % python3
Python 3.9.6 (default, May 22 2026, 11:13:45)
[Clang 21.0.0 (clang-2100.1.1.101)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
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29736
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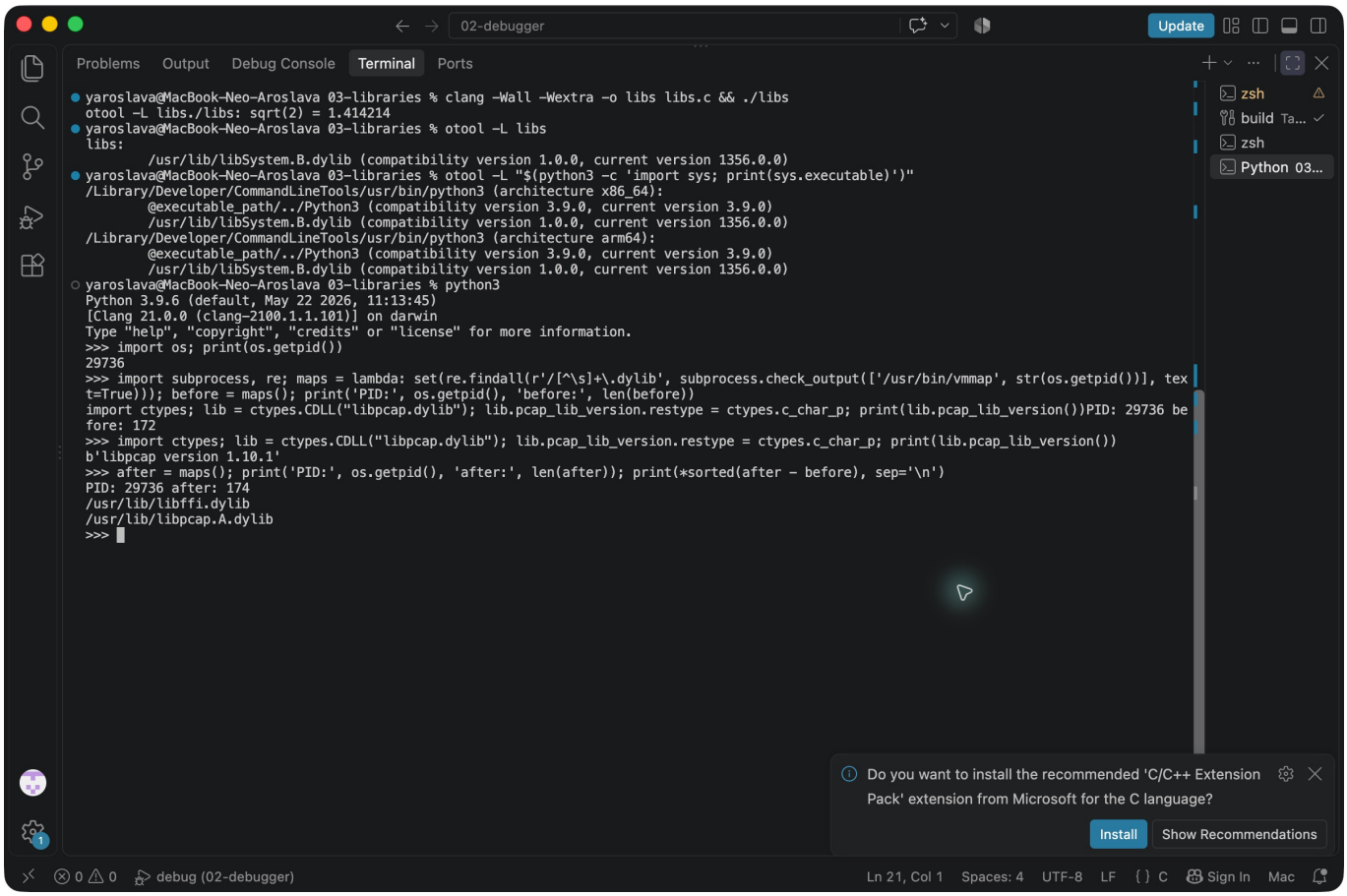
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b'libpcap version 1.10.1'
>>> after = maps(); print('PID:', os.getpid(), 'after:', len(after)); print(*sorted(after - before), sep='\n')
PID: 29736 after: 174
/usr/lib/libffi.dylib
/usr/lib/libpcap.A.dylib
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